

Theoretical Analysis of Kriging Error under Training Data Subsampling

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Abstract

Gaussian process regression, or kriging, provides an optimal linear estimator under Gaussian prior assumptions. However, its high computational cost limits its applicability to large datasets. A common workaround involves subsampling the training data, typically by retaining only points near the estimation location.

This induces a loss of accuracy whose impact is often left unquantified. In this paper, we derive explicit bounds on the kriging error introduced by such data reduction strategies. By expressing the variance and interpolation errors in terms of residual covariances and observations, we construct entrywise and worst-case upper bounds. We further explore the behaviour of these bounds under varying regimes of the observation covariance matrix's spectral properties to determine kriging robustness under subsampling. These results bridge the gap between heuristic subsampling strategies and rigorous performance guarantees, providing computable criteria for robust subsampled Gaussian process regression.

Keywords: Gaussian process, kriging, subsampling, error bounds, computational complexity reduction

1. Introduction

Gaussian process regression (GPR) [1] is a nonparametric Bayesian method widely used in spatial statistics such as astrophysics and geostatistics, and surrogate modelling. It is valued for its ability to provide not only point estimates but also uncertainty quantifications. However, its computational cost scales cubically with the number of training points, making it prohibitive in large-scale or real-time applications.

A substantial body of work has addressed this limitation through approximation techniques. Low-rank methods, such as the Nyström approximation [2] or the Karhunen–Loève decomposition [3] can be used to reduce the computational cost. Sparse variational methods can also approximate the covariance structure by using pseudo-inputs [4] or spectral expansions [5]. While these methods are effective, they typically alter the underlying Gaussian process model, potentially affecting its interpretability or calibration.

An alternative strategy is to reduce complexity by directly subsampling the training data. Local selections only consider the closest points to the prediction location [6, 7], while other methods include submodularity [8], greedy heuristics [9], entropy-based criteria [10], determinantal point processes [11], or nearest-neighbour techniques [12]. While such approaches are computationally appealing and preserve the original model assumptions, they usually lack rigorous guarantees on the error induced by discarding observations. Some theoretical efforts have examined error bounds for GPR itself [13], low-rank approximations [14] or random subsampling [15], but general bounds quantifying the training data subsampling remain scarce.

In this work, we provide explicit theoretical error bounds for subsampled GPR. Unlike previous methods that alter the model or rely on heuristic selection, our analysis focuses directly on the Gaussian process structure and yields interpretable, computable metrics to assess the impact of subsampling. In section 2, we recall the standard setting of GPR and introduce error metrics for subsampling. Section 3 reformulates both the L^2 variance error and the L^1 interpolation error in terms of residual quantities, representing the information contained in the training data that is lost during subsampling. This leads to closed-form expressions that characterise the loss of information from discarding data. Building on these results, section 4 derives both entrywise and worst-case upper bounds, expressed through spectral properties of the observation covariance matrix. Together, these results clarify how the robustness to subsampling depends on the eigenvalues of the latter, the spatial distribution of training points, and the observation noise regime.

2. Problem Statement

2.1. Gaussian Process Regression

Let $f : \Omega \rightarrow \mathbb{R}$ be a realisation of a zero-mean Gaussian process of covariance kernel $k : \Omega \times \Omega \rightarrow \mathbb{R}$. For $\tilde{X} \triangleq [\tilde{x}_1 \ \dots \ \tilde{x}_{\tilde{n}}]^T \in \Omega^{\tilde{n}}$ a set of $\tilde{n}$ training data, we are given the noisy realisations $\forall i \in \llbracket 1, \tilde{n} \rrbracket$, $\tilde{z}_i \triangleq f(\tilde{x}_i) + \eta_i$ with $\eta_i \sim \mathcal{N}(0, \tilde{r})$. We also define $\tilde{Z} \triangleq [\tilde{z}_1 \ \dots \ \tilde{z}_{\tilde{n}}]^T \in \mathbb{R}^{\tilde{n}}$. The kriging estimator $f_{\tilde{X}}^* : \Omega \rightarrow \mathbb{R}$ is the function that associates the best linear unbiased estimator (BLUE) at each point of Ω with respect to $\{\tilde{X}, \tilde{Z}\}$. Formally, the BLUE is defined as the linear combination of $\tilde{Z}$ that minimises the error covariance:

$$\forall x \in \Omega, f_{\tilde{X}}^*(x) \triangleq \Lambda(x, \tilde{X})^T \tilde{Z} \quad (1)$$

$$\text{with } \Lambda(x, \tilde{X}) = \arg \min(\mathbb{V}(f(x) - f_{\tilde{X}}^*(x))) \quad (2)$$

As proven by [1], the optimum is reached when $f_{\tilde{X}}^*(x)$ corresponds to the *a posteriori* expectation of x conditional to $\{\tilde{X}, \tilde{Z}\}$:²

$$f_{\tilde{X}}^* : x \in \Omega \mapsto \mathbb{E}[f(x) | \tilde{Z}] = k(x, \tilde{X})(k(\tilde{X}, \tilde{X}) + \tilde{r}I_{\tilde{n}})^{-1} \tilde{Z} \quad (3)$$

The associated error variance is:

$$r_{\tilde{X}}^* : x \in \Omega \mapsto \mathbb{V}(f_{\tilde{X}}^*(x) - f(x)) = k(x, x) - k(x, \tilde{X})(k(\tilde{X}, \tilde{X}) + \tilde{r}I_{\tilde{n}})^{-1} k(\tilde{X}, x) \quad (4)$$

A major limitation of this method is the $\mathcal{O}(\tilde{n}^3)$ computational cost, stemming from the inversion of the Gram matrix $G(\tilde{X}) \triangleq k(\tilde{X}, \tilde{X}) + \tilde{r}I_{\tilde{n}}$ in both (3) and (4). One common approach to mitigate this is to omit certain elements of $\tilde{X}$ to reduce the size of $G(\tilde{X})$. However, this approximation introduces a loss of accuracy in (2). This paper aims to quantify the error induced by such subsampling approximations.

2.2. Subsampling Error Metrics

Formally, subsampling is achieved by splitting $\tilde{X}$ into $[\tilde{X}_c \ \tilde{X}_f]$, with $\tilde{X}_c$ being the $\tilde{n}_c$ retained data, usually close to the considered kriging location $x^* \in \Omega$, and $\tilde{X}_f$ being the $\tilde{n}_f$ discarded data, usually far from x^* . A similar operation is performed on $\tilde{Z}$. This decomposition provides the foundation for all subsequent error bounds, as it explicitly isolates the predictive contribution of the discarded points from that of the retained training set.

Two error metrics between $f_{\tilde{X}}^*(x^*)$ and $f_{\tilde{X}_c}^*(x^*)$ are considered in this paper, each capturing a distinct aspect of the kriging approximation. The first metric corresponds to the root mean squared error, and is formally introduced in Definition 1. Conditional to $\tilde{Z}$, the kriging estimator is a deterministic function, , a natural alternative is to measure the absolute difference between the two estimators, as proposed by Definition 2.

Definition 1. Let e_r denote the (root) variance error between $f_{\tilde{X}}^*$ and $f_{\tilde{X}_c}^*$ in the L^2 space:

$$e_r : x \in \Omega \mapsto \|f_{\tilde{X}}^*(x) - f_{\tilde{X}_c}^*(x)\|_{L^2} = \sqrt{\mathbb{E}[(f_{\tilde{X}}^*(x) - f_{\tilde{X}_c}^*(x))^2]}$$

Definition 2. The L^1 interpolation error e_f between $f_{\tilde{X}}^*$ and $f_{\tilde{X}_c}^*$ is defined as:

$$e_f : x \in \Omega \mapsto \|f_{\tilde{X}}^*(x) - f_{\tilde{X}_c}^*(x)\|_{L^1} = |f_{\tilde{X}}^*(x) - f_{\tilde{X}_c}^*(x)|$$

3. Data Selection Errors

3.1. Importance of the Gram Matrix

As stated in section 2.1, most of the computational cost comes from inverting the Gram matrix. Using the decomposition introduced in section 2.2 leads to the following block partitioning of $G(\tilde{X})$:

$$G(\tilde{X}) = \begin{bmatrix} A & B \\ B^T & D \end{bmatrix} = \begin{bmatrix} G(\tilde{X}_c) & k(\tilde{X}_c, \tilde{X}_f) \\ k(\tilde{X}_c, \tilde{X}_f)^T & G(\tilde{X}_f) \end{bmatrix} \quad (5)$$

Let $G_h(\tilde{X}_c)$ be the $\tilde{n} \times \tilde{n}$ hollowed-out Gram matrix $G_h(\tilde{X}_c) \triangleq \begin{bmatrix} G(\tilde{X}_c) & 0_{\tilde{n}_c, \tilde{n}_f} \\ 0_{\tilde{n}_f, \tilde{n}_c} & 0_{\tilde{n}_f, \tilde{n}_f} \end{bmatrix}$. Proposition 1 states that this matrix acts as a substitute for $G(\tilde{X})$ in the subsampled case.

Proposition 1. *The subsampled kriging estimator verifies:*

$$f_{\tilde{X}_c}^*(x^*) = k(x^*, \tilde{X})G_h(\tilde{X}_c)^\dagger \tilde{Z}; \quad r_{\tilde{X}_c}^*(x^*) = k(x^*, x^*) - k(x^*, \tilde{X})G_h(\tilde{X}_c)^\dagger k(\tilde{X}, x^*)$$

where $\dagger$ denotes the Moore-Penrose pseudo-inverse symbol.

Proof. The result comes from applying (3) and (4) to $\tilde{X}_c$, and noting that $G_h(\tilde{X}_c)^\dagger = \begin{bmatrix} G(\tilde{X}_c)^{-1} & 0 \\ 0 & 0 \end{bmatrix}$. $\square$

Although it is not directly used in computations, Proposition 1 further highlights the importance of the Gram matrix in the subsampling problem. In particular, Definition 3 serves as the foundation for the key reformulations of the studied errors established by Proposition 2.

Definition 3. *The Gram inverse difference matrix $\mathfrak{D}_G$ is the quantity defined as:*

$$\mathfrak{D}_G \triangleq G(\tilde{X})^{-1} - G_h(\tilde{X}_c)^\dagger$$

Proposition 2. *The two subsampling errors can be respectively expressed as:*

$$e_r(x^*) = \sqrt{k(x^*, \tilde{X}) \mathfrak{D}_G k(\tilde{X}, x^*)}; \quad e_f(x^*) = |k(x^*, \tilde{X}) \mathfrak{D}_G \tilde{Z}|$$

Proof. Since the kriging interpolator is equal to the *a posteriori* expectation of $f(x)$ knowing the training data, it can be seen as the orthogonal projection of $f(x)$ onto the subspace spanned by the noisy observations. The difference between the two interpolators can be decomposed as $f_{\tilde{X}}^*(x^*) - f_{\tilde{X}_c}^*(x^*) = e_{\tilde{X}_c} - e_{\tilde{X}}$, where $e_{\tilde{X}_c} \triangleq f(x^*) - f_{\tilde{X}_c}^*(x^*)$ and $e_{\tilde{X}} \triangleq f(x^*) - f_{\tilde{X}}^*(x^*)$. Given that $e_{\tilde{X}}$ is the error after projection onto the space generated by all of $\tilde{Z}$, it is orthogonal to $f_{\tilde{X}}^*(x^*)$. Since $f_{\tilde{X}_c}^*(x^*)$ also belongs to this space, it comes that $e_{\tilde{X}}$ is orthogonal to $f_{\tilde{X}}^*(x^*) - f_{\tilde{X}_c}^*(x^*)$. Therefore $e_r(x^*)^2 = \mathbb{E}[e_{\tilde{X}_c}^2] - \mathbb{E}[e_{\tilde{X}}^2] = r_{\tilde{X}_c}^*(x^*) - r_{\tilde{X}}^*(x^*)$. Proposition 1 concludes the demonstration for both $e_r(x^*)$ and $e_f(x^*)$. $\square$

3.2. Block Inversion and Residual Quantities

To isolate the impact of discarding $\tilde{X}_f$ on the kriging estimator, $\mathfrak{D}_G$ is decomposed using the Woodbury block matrix inversion formula [16]:

$$\mathfrak{D}_G = \begin{bmatrix} A & B \\ B^T & D \end{bmatrix}^{-1} - \begin{bmatrix} A^{-1} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} -A^{-1}B \\ I_{\tilde{n}_f} \end{bmatrix} S^{-1} \begin{bmatrix} -A^{-1}B \\ I_{\tilde{n}_f} \end{bmatrix}^T \quad (6)$$

where $S \triangleq D - B^T A^{-1} B$ is the Schur complement of $G(\tilde{X})$ with respect to $A = G(\tilde{X}_c)$. This decomposition, combined with Proposition 2, reveals two residual quantities that contribute to the error introduced by subsampling. The first one, called residual covariance function, quantifies the portion of the covariance between x^* and the discarded training set $\tilde{X}_f$ not accounted for by the retained set $\tilde{X}_c$. Similarly, the residual observation vector captures the information contained in the discarded observations $\tilde{Z}_f$ and unexplained by the retained observations $\tilde{Z}_c$. Both expressions are given by Definition 4.

Definition 4. *The residual covariance function φ and the residual observation vector ψ are respectively defined as:*

$$\varphi : x \in \Omega \mapsto k(\tilde{X}_f, x) - k(\tilde{X}_f, \tilde{X}_c)G(\tilde{X}_c)^{-1}k(\tilde{X}_c, x); \quad \psi \triangleq \tilde{Z}_f - k(\tilde{X}_f, \tilde{X}_c)G(\tilde{X}_c)^{-1}\tilde{Z}_c$$

Lemma 1 gives relevant identities that link (6) with φ and ψ . Proposition 3 and Corollary 3.1 push this further by showing that the studied metrics can be fully expressed as functions of those residual quantities.

Lemma 1. *The following residual identities hold:*

$$\varphi(x^*) = \begin{bmatrix} -A^{-1}B \\ I_{\tilde{n}_f} \end{bmatrix}^T k(\tilde{X}, x^*); \quad \varphi(\tilde{X}_f) = S - \tilde{r}I_{\tilde{n}_f}; \quad \psi = \begin{bmatrix} -A^{-1}B \\ I_{\tilde{n}_f} \end{bmatrix}^T \tilde{Z}$$

Proof. Direct calculations give the desired results. $\square$

Proposition 3. *The residual forms of the two errors are given below:*

$$e_r(x^*) = \sqrt{\varphi(x^*)^T (\varphi(\tilde{X}_f) + \tilde{r}I_{\tilde{n}_f})^{-1} \varphi(x^*)}; \quad e_f(x^*) = |\varphi(x^*)^T (\varphi(\tilde{X}_f) + \tilde{r}I_{\tilde{n}_f})^{-1} \psi|$$

Proof. These expressions are obtained by applying (6) and Lemma 1 to Proposition 2. $\square$

Corollary 3.1. *Let $\langle \cdot | \cdot \rangle_S$ be the following Mahalanobis inner product: $\forall u, v \in (\mathbb{R}^{\tilde{n}_f})^2$, $\langle u | v \rangle_S \triangleq u^T S^{-1} v$, and $\|\cdot\|_S$ the associated norm. The kriging errors can be interpreted as metrics in the $(\mathbb{R}^{\tilde{n}_f}, \langle \cdot | \cdot \rangle_S)$ euclidean space:*

$$e_r(x^*) = \|\varphi(x^*)\|_S; \quad e_f(x^*) = |\langle \varphi(x^*) | \psi \rangle_S|$$

4. Error Bounds and Asymptotic Behaviour

4.1. Entrywise Upper Bounds

The goal of this section is to derive upper bounds for $e_r(x^*)$ and $e_f(x^*)$ from Corollary 3.1. For $m, n \in \mathbb{N}^2$, let $J_{m,n}$ denote the $m \times n$ matrix full of 1. We assume the following entrywise bounds:

$$\begin{aligned} |k(\tilde{X}_f, \tilde{X}_c)| &\leq \alpha J_{\tilde{n}_f, \tilde{n}_c}; \quad |k(\tilde{X}_j, \tilde{X}_j)| \leq (k_{\max} - \kappa) I_{\tilde{n}_j} + \kappa J_{\tilde{n}_j, \tilde{n}_j} \\ |k(\tilde{X}_c, x^*)| &\leq \kappa J_{\tilde{n}_c, 1}; \quad |k(\tilde{X}_f, x^*)| \leq \varepsilon J_{\tilde{n}_f, 1}; \quad |\tilde{Z}_j| \leq \tilde{z}_{\max} J_{\tilde{n}_j, 1} \end{aligned} \quad (7)$$

with $j \in \{c, f\}$. Here, κ , α , and ε bound respectively the intra-set covariance within $\tilde{X}_j$, the cross covariance between $\tilde{X}_c$ and $\tilde{X}_f$, and the covariance between x^* and $\tilde{X}_f$. The constants $k_{\max}$ and $\tilde{z}_{\max}$ respectively denote the maximum values of k and $\tilde{z}_i$, $i \in \llbracket 1, \tilde{n} \rrbracket$. Ideally, intra-set covariances are strong, while cross covariances with $\tilde{X}_f$ remain small: $0 < \varepsilon$, $\alpha \ll \kappa \leq k_{\max}$. These assumptions are consistent with the subsampling strategy described in section 2.2, where the elements of $\tilde{X}_c$ are chosen near x^* , while those from $\tilde{X}_f$ are farther away.

Let $\|\cdot\|_2$ and $\|\cdot\|_{op}$ denote respectively the euclidean vector norm and the associated matrix operator norm. We define $\lambda_j^- \triangleq \lambda_{\min}(G(\tilde{X}_j))$, $j \in \{c, f\}$. Lemma 2 and Proposition 4 respectively give lower bounds for the residual quantities and the two errors.

Lemma 2. *Assuming k verifies (7), the following identities hold:*

$$\|\varphi(x^*)\|_2 \leq \sqrt{\tilde{n}_f} \frac{\lambda_c^- \varepsilon + \alpha \tilde{n}_c \kappa}{\lambda_c^-}; \quad \|S^{-1}\|_{op} \leq \frac{\lambda_c^-}{\lambda_c^- \lambda_f^- - \alpha^2 \tilde{n}_c \tilde{n}_f}; \quad \|\psi\|_2 \leq \sqrt{\tilde{n}_f} \frac{\lambda_c^- + \alpha \tilde{n}_c}{\lambda_c^-} \tilde{z}_{\max}$$

Proof. The results for $\|\varphi(x^*)\|_2$ and $\|\psi\|_2$ can be directly obtained using the fact that $G(\tilde{X}_c)$ is symmetric definite positive. Let u be a unit eigenvector associated with λ_f^- . Since $S = \varphi(\tilde{X}_f) + \tilde{r}I_{\tilde{n}_f}$ is also symmetric definite positive, applying the Rayleigh quotient [17] to S and u yields the desired formula for $\|S^{-1}\|_{op}$. $\square$

Proposition 4. *Under the same assumptions as Lemma 2, the ensuing inequalities are satisfied:*

$$e_r(x^*) \leq \frac{\sqrt{\tilde{n}_f} (\lambda_c^- \varepsilon + \alpha \tilde{n}_c \kappa)}{\sqrt{\lambda_c^- (\lambda_c^- \lambda_f^- - \alpha^2 \tilde{n}_c \tilde{n}_f)}}; \quad e_f(x^*) \leq \frac{\tilde{n}_f (\lambda_c^- \varepsilon + \alpha \tilde{n}_c \kappa) (\lambda_c^- + \alpha \tilde{n}_c)}{\lambda_c^- (\lambda_c^- \lambda_f^- - \alpha^2 \tilde{n}_c \tilde{n}_f)} \tilde{z}_{\max}$$

Proof. For $u, v \in (\mathbb{R}^{\tilde{n}_f})^2$, the inner product verifies the Cauchy-Schwarz inequality [17]: $|\langle u | v \rangle_S| \leq \|u\|_2 \|v\|_2 \|S^{-1}\|_{op}^{-1}$. Applying the inequalities from Lemma 2 concludes this demonstration. $\square$

4.2. Worst-Case Bounds

Unfortunately, the bounds from Proposition 4 cannot be further tightened in their current form. While [18] shows that λ_j^- cannot collapse to 0, the bounds remain loose without additional information. To address this issue, we assume in this section that the bounds from (7) are attained. For the sake of simplicity, the data within $\tilde{X}_c \cup \{x^*\}$ and $\tilde{X}_f$ are assumed to be positively correlated, meaning that the bounds for κ are positive, while the rest behaves as either their plus or minus bound.

In the rest of this article, every quantity is given a worst-case counterpart, denoted by the w symbol. Worst-case quantities are defined by replacing k with the relevant bound from (7) in their base form. Applying the residual identities and Corollary 3.1 to the two worst-case error bounds gives:

$$e_r^w(x^*) = \|\varphi^w(x^*)\|_{S^w}; \quad e_f^w(x^*) = |\langle \varphi^w(x^*) | \psi^w \rangle_{S^w}| \quad (8)$$

Let $\lambda_j^{+w} \triangleq \lambda_{\max}(G^w(\tilde{X}_j))$, $j \in \{c, f\}$. Since $G^w(\tilde{X}_j)$ is known, its spectrum can be fully determined. In particular $\lambda_j^{+w} = k_{\max} + (\tilde{n}_j - 1)\kappa + \tilde{r}$. Lemma 3 and Proposition 5 describe the worst-case behaviour of the residual quantities and the errors respectively.

Lemma 3. *The worst-case residual function and vector verify:*

$$(S^w)^{-1} = \frac{1}{\lambda_c^{+w} - \tilde{n}_c \kappa} \left(I_{\tilde{n}_f} - \frac{\lambda_c^{+w} \kappa - \alpha^2 \tilde{n}_c}{\lambda_c^{+w} \lambda_f^{+w} - \alpha^2 \tilde{n}_c \tilde{n}_f} J_{\tilde{n}_f, \tilde{n}_f} \right)$$

$$\varphi^w(x^*) = \frac{\pm \lambda_c^{+w} \varepsilon \pm \alpha \tilde{n}_c \kappa}{\lambda_c^{+w}} J_{\tilde{n}_f, 1}; \quad \psi^w = \frac{\pm \lambda_c^{+w} \pm \alpha \tilde{n}_c}{\lambda_c^{+w}} \tilde{z}_{\max} J_{\tilde{n}_f, 1}$$

Proof. In the worst case, $B^T A^{-1}$ becomes $\frac{\alpha}{\lambda_c^{+w}} J_{\tilde{n}_f, \tilde{n}_c}$. The desired expressions can be derived from this result and the Woodbury matrix inversion formula [16]. $\square$

Proposition 5. *The worst-case bounds behave as described below:*

$$e_r^w(x^*) = \frac{\sqrt{\tilde{n}_f} |\lambda_c^{+w} \varepsilon \pm \alpha \tilde{n}_c \kappa|}{\sqrt{\lambda_c^{+w} (\lambda_c^{+w} \lambda_f^{+w} - \alpha^2 \tilde{n}_c \tilde{n}_f)}}; \quad e_f^w(x^*) = \frac{\tilde{n}_f |\lambda_c^{+w} \varepsilon \pm \alpha \tilde{n}_c \kappa| |\lambda_c^{+w} \pm \alpha \tilde{n}_c|}{\lambda_c^{+w} (\lambda_c^{+w} \lambda_f^{+w} - \alpha^2 \tilde{n}_c \tilde{n}_f)} \tilde{z}_{\max}$$

Proof. We precompute $q \triangleq \langle J_{\tilde{n}_f, 1} | J_{\tilde{n}_f, 1} \rangle_{S^w} = \frac{\tilde{n}_f \lambda_c^{+w}}{\lambda_c^{+w} \lambda_f^{+w} - \alpha^2 \tilde{n}_c \tilde{n}_f}$. The announced forms are obtained by bilinearity of the inner product from (8) with q and Lemma 3. $\square$

While the expressions in Proposition 4 and Proposition 5 are extremely similar, the latter presents tighter bounds, as they circumvent the triangular inequality and use the maximum eigenvalues of the Gram matrices instead of their minimum.

4.3. Eigenvalue-Dependent Limits

Since Proposition 5 explicitly relates the error bounds to the spectrum of the Gram matrices, it is natural to examine their behaviour under two asymptotic eigenvalue regimes. The first case arises when $\tilde{r} \gg \tilde{n}_j k_{\max}$, and corresponds to an extreme noise-dominated setting in which almost no information can be extracted. This regime is mainly analysed to confirm the consistency of the bounds. Conversely, when $\tilde{r}$ is small or $\tilde{n}_j \rightarrow +\infty$, and as $\kappa \rightarrow k_{\max}$, the kriging estimators provide reliable approximations of $f(x^*)$. In terms of eigenvalues, this amounts to assuming the following limits:

$$\lambda_j^{+w} \sim \begin{cases} \tilde{r} \rightarrow +\infty, & \text{degenerate case} \\ \tilde{n}_j k_{\max} \rightarrow +\infty, & \text{general case} \end{cases} \quad (9)$$

Propositions 6 and 7 give simple asymptotic equivalents of the two errors for each eigenvalue regime.

Proposition 6. *In the degenerate case, the variance and interpolation errors evolve as follows:*

$$e_r^w(x^*) \sim \sqrt{\frac{\tilde{n}_f}{\tilde{r}}} \varepsilon; \quad e_f^w(x^*) \sim \frac{\tilde{n}_f}{\tilde{r}} \varepsilon \tilde{z}_{\max}$$

Proposition 7. *The two errors present the following behaviour under the general regime:*

$$e_r^w(x^*) \sim \sqrt{k_{\max}} \frac{|\varepsilon \pm \alpha|}{\sqrt{k_{\max}^2 - \alpha^2}}; \quad e_f^w(x^*) \sim \frac{|\varepsilon \pm \alpha| |k_{\max} \pm \alpha|}{k_{\max}^2 - \alpha^2} \tilde{z}_{\max}$$

Proof. Both Propositions 6 and 7 are obtained by plugging (9) into Proposition 5. $\square$

Unsurprisingly, the asymptotic bounds tend to 0 when the observation noise dominates, as both estimators become equally inaccurate. In the general case however, two factors drive the asymptotic behaviours: a denominator involving $k_{\max}^2 - \alpha^2$, which diverges as α increases, and a numerator containing $|\varepsilon \pm \alpha|$, which drives both errors to 0 when ε and α tend to 0. As suggested in section 4.1, this implies that α must remain small for $\tilde{X}_f$ to be safely discarded.

For $e_f^w(x^*)$, the observations themselves become decisive. When the numerator is $|\varepsilon \pm \alpha| |k_{\max} - \alpha|$, both subsets typically share the same $\tilde{z}_{\max}$ value. Discarding $\tilde{X}_f$ therefore removes no information, and the denominator reduces to $k_{\max} + \alpha$. In contrast, the form in $|\varepsilon \pm \alpha| |k_{\max} + \alpha|$ occurs when $\tilde{Z}_c$ and $\tilde{Z}_f$ are so drastically different that discarding a part of a strongly correlated and informative subset causes the error to scale as $\mathcal{O}(1/(k_{\max} - \alpha))$.

5. Conclusions

This paper derived explicit L^1 and L^2 error bounds for kriging under training data subsampling. By decomposing the Gramian covariance matrix of the observations and introducing residual quantities, we reformulated the subsampling errors into interpretable components. This enabled the derivation of closed-form expressions for entry-wise and worst-case bounds, as well as worst-case equivalents that directly relate the error behaviour to the spectral properties of the covariance matrices of the retained/discarded data. Our analysis highlights how robustness to subsampling jointly depends on the Gram matrix spectral properties and the spatial cross-covariance between the discarded and the retained training points.

Beyond clarifying the theoretical impact of subsampling, these results provide practical criteria for evaluating subsampling strategies, offering a principled alternative to heuristic selection rules. In particular, they make it possible to link the tolerated prediction error with the number of training points that can be discarded: if a maximum error level is specified, the bounds indicate how much data reduction is safe; conversely, given a subsampling strategy, they provide a quantifiable estimate of the induced error.

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